SAMUVEL NATARAJAN

DevOps Engineer | Azure & AKS | CI/CD | Infrastructure as Code | Kubernetes

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Professional Summary

DevOps Engineer with 3.5 years of experience in cloud infrastructure, CI/CD automation, and containerized software delivery. Hands-on with Azure (AKS, Virtual Machines, VNet, DNS Zones, Storage Accounts, Key Vault, Logic Apps) alongside AWS, using Docker, Kubernetes, Terraform, Jenkins, GitHub Actions, and Argo CD across development, staging, and production environments. Experienced in monitoring with Prometheus, Grafana, and Loki, integrating security scanning (Trivy, SonarQube, Gitleaks, OWASP Dependency Check) into pipelines, and troubleshooting deployment and pipeline issues in Agile teams.

Technical Skills

Microsoft Azure: Microsoft Entra ID, Azure RBAC, AKS, Virtual Machines, VNet, DNS Zones, Storage Accounts, Key Vault, Logic Apps, Azure Firewall and NSGs

CI/CD and Version Control: Jenkins, GitHub Actions, Argo CD (GitOps), Git, GitHub, automated build, test, security scan, and deployment pipelines

Infrastructure as Code: Terraform, reusable modules, remote state management, multi-environment provisioning

Containers and Orchestration: Docker, Kubernetes, Azure AKS, Amazon EKS, Helm, HPA

Scripting and Automation: Python, Bash/Shell, YAML, Groovy

Monitoring and Observability: Prometheus, Grafana, Loki, Alertmanager, Slack and email alerts

Security and Networking: VNet/VPC, DNS, load balancing, Kubernetes Ingress, SSL/TLS, IAM, Trivy, SonarQube, Gitleaks, OWASP Dependency Check

AWS: EC2, EKS, ECR, S3, VPC, IAM, ALB, Route 53, KMS

Databases and Tools: PostgreSQL, MongoDB, Pytest, Selenium, Agile/Scrum

Currently Learning (Azure): Azure DevOps and Azure Pipelines, Azure Monitor, Application Insights, Log Analytics

Professional Experience

DevOps Engineer | Majordomo Systems India Private Limited **Jul 2025 – Present**
Chennai, India

- **Kubernetes and Containers (AKS/EKS):** Deployed and operated containerized workloads on Azure AKS and AWS EKS using Docker, Helm, and kubectl across development, staging, and production environments, including Kubernetes networking and ingress.
- **CI/CD and DevSecOps:** Built Jenkins CI/CD pipelines triggered by code changes to automate builds, testing, security scans, and deployments. Integrated Trivy, SonarQube, Gitleaks, and OWASP Dependency Check, introducing Gitleaks and OWASP Dependency Check as the organization's first implementation.
- **GitOps:** Managed Argo CD deployments from Git-based Kubernetes manifests and Helm configurations in non-production environments.
- **Cloud Infrastructure and IaC:** Provisioned and managed AWS infrastructure using reusable Terraform modules across development, staging, and production, including VPCs, EKS, and EC2.
- **Monitoring and Operations:** Used Prometheus and Grafana for metrics and dashboards, Loki for centralized logging, and Slack/email alerting. Administered Linux workloads, troubleshot operational issues, and automated recurring tasks with Bash and Python.
- **Testing Automation:** Developed Python test suites using Pytest and Selenium for API, regression, and UI testing, integrating test execution into CI/CD pipelines.
- **Cost Optimization:** Improved cloud cost visibility through non-production resource scheduling, resource tagging, instance rightsizing, usage reviews, and AWS budget alerts.
- **AI-Agent Tooling and Backend Development:** Contributed to internal AI-agent tooling, built MCP (Model Context Protocol) tools, developed and tested Go backend features, and evaluated LLM outputs using an LLM-as-a-judge approach.
- **PaaS and SaaS Operations:** Supported cloud infrastructure and software delivery across PaaS and SaaS product environments.

Tailwinds Cloud Services Private Limited **Apr 2023 – Jun 2025**
Chennai, India

DevOps Engineer *Jul 2023 – Jun 2025*

- Supported secure delivery of a PaaS product, maintaining CI/CD environments and Azure and AWS infrastructure.

Client: Payment Gateway Company *Docker, Kubernetes, Kong, MongoDB, Jenkins, Argo CD*

- Migrated Docker-based application services to Kubernetes on AWS and introduced Kong for internal service routing.
- Automated application deployments using Jenkins and Argo CD, improving release consistency and operational reliability.
- Supported cost optimization and infrastructure monitoring through proactive maintenance and automation.

Client: IIT Madras Research Park *AWS, WordPress*

- Deployed and maintained WordPress websites on AWS, managing provisioning, scaling, patching, and updates.
- Monitored infrastructure health and applied maintenance updates, contributing to reported service uptime.

Software Engineer Intern *Apr 2023 – Jun 2023*

Projects

Cloud-Native DevOps Platform – Personal Project GitHub Repository

- Built a cloud-native microservices platform using Node.js/Express, Python/FastAPI, PostgreSQL, Docker, and Kubernetes.
- Developed modular Terraform infrastructure for AWS VPC, IAM, and EKS, including public/private subnets, NAT Gateway, and managed node groups.
- Implemented GitHub Actions workflows for automated testing, container builds, and security checks using Trivy, Gitleaks, and Hadolint.
- Created multi-stage, non-root Docker builds, Kubernetes health probes, resource requests and limits, and environment-parameterized Helm deployments.
- Automated server configuration with Ansible and kept credentials and Terraform state out of source control.

Certification

AWS Certified Cloud Practitioner Amazon Web Services (AWS), 2023

Education

Bachelor of Computer Applications (BCA) Expected 2027
Tamil Nadu Open University (TNOU)
Distance education pursued alongside professional experience.